

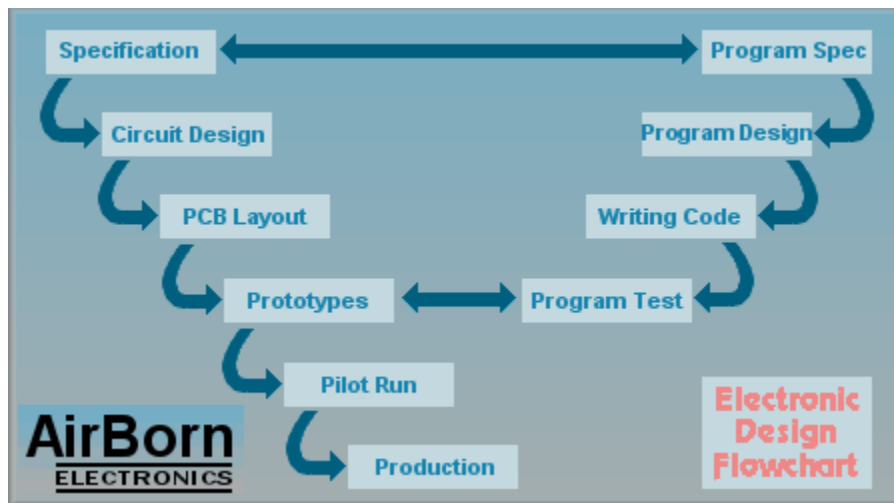


Electronics Design



When designing any product, the **first steps** are to **identify the need** the product will fill, and **identify the market** it will be sold into. Then we consider the competing products that can already perform the job - *even 'unique' products have some kind of competition, somewhere*. Finally we build up a specification for a new product positioned successfully in the market. This is detailed a little more in our guide to [writing a new product spec](#).

This guide describes our process for designing a new electronic circuit board, typically one that is microprocessor based with embedded firmware. This same process is used for most products we design. Usually an industrial designer is employed to produce the product enclosure or case, with labelling and graphics. We deliver working prototypes and [technical documentation](#) to the client - and then the product goes on to beta testing, pilot run, and production release.



Nothing is particularly hard if you divide it into small jobs. --
Henry Ford

The Specification

The Specification is the customer's instructions to the designer describing *the features the new product should have*. (See: [More info - Specifications](#)) A specification starts out as ideas and a wish list, then gradually gets nailed down to something more formal and explicit. It is a good idea to have a list of "*reasons why*" and "*how it's done*" to explain the specification, but keep these separate from the main document.

We have, online, a [guide to writing a specification](#) for a new electronic product, help with [specification construction](#), [examples of specifications](#), and a [glossary of technical information](#).



The basic, introduction explanation of the products function within a specification should be quite short, keep it to one page:

One Page Principle: A specification that will not fit on one page of 8.5x11 inch paper cannot be understood. -- Mark Ardis 😊

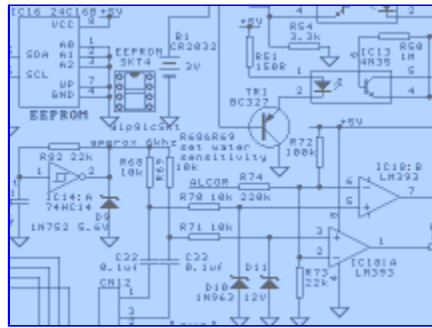

[Contact](#)

[Detail: Specs](#)
[Example](#)
[Specs](#)


Where you see the magnifying glass symbol you can **CLICK** to get **MORE DETAIL**

The Circuit Design

The **Circuit Diagram**, also called the **Schematic** or **Logic Diagram**, maps out the electronics and connections in the most readily readable form. (See: [More info - Circuit Design](#)) The designer needs to do background work while producing the Circuit diagram, researching specifications of components, interaction between components (especially timing and loading) physical packages, and arrangement of connector pinouts. The circuit will often start on paper and finish in Computer Aided Design (CAD) format.

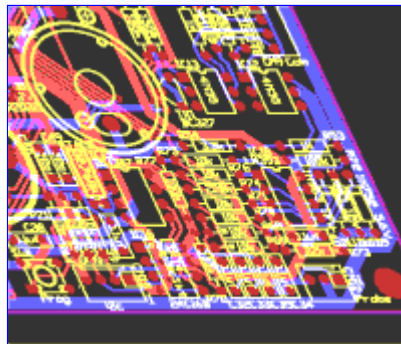


The finished circuit diagram, supported by notes if required, is the main reference document for the design.


[Detail: Circuit Design](#)
[Example](#)
[Circuit](#)

The Printed Circuit Board Layout

The Printed Circuit Board (PCB) is the laminate to which all the electronic components are soldered, with one or more layers of etched metal tracks making the connections. (See: [More info - PCB Layout](#)) The components and connections in the PCB Layout are derived from the circuit diagram, and physically placed and routed by the designer to get the best result. The PCB Layout defines the **final physical form of the circuit**, and enclosure and labelling details can be finalised as the layout is completed.



When the PCB layout is complete, the final CAD file is sent to a subcontract PCB laminate manufacturer. (for detail on PCB laminate production see [More info - PCB Etching](#)) The manufacturer returns the etched PCB laminates a few days

later, ready for assembly into prototypes.

Hofstadter's Law: It always takes longer than you expect, even when you take Hofstadter's Law into account. 😊



[Detail: Layout](#)

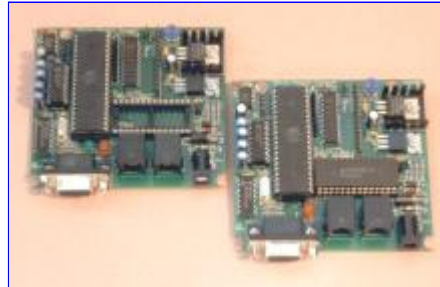


[Detail: Etch](#)

Example

Prototypes

At **AirBorn Electronics** we usually make 2 or 3 identical prototypes at the same time. (See: [More info - Prototypes](#)) The Prototype circuitry is debugged in stages. The debugging proceeds according to the [debug test procedure](#), which is written for the product as it is designed. Prototype microprocessor circuits are generally debugged with specially written diagnostic code, again progressing in stages.



When confidence has been gained in the operation of the prototype hardware, debug of the actual prototype program ("Firmware") can begin, on the target.

Firmware

Software for the electronics is often 50% or more of the design work in a project. The programming that controls the microprocessor usually gives the electronic product its *features* and *"glitz"* whereas the hardware is the purely functional side of the equation.

The programming or coding for electronic products is usually written in 'C' or assembler, whereas the programming that runs on a PC will normally be written in C++, java or another language (and almost certainly not assembler). The programming for microcontrollers is totally different from that for a PC in any case - it is usually "more detailed" and less "generic" than the type of programming made for a personal computer, because it is more tightly coupled to the electronic hardware on which it runs, and because it usually has to respond more quickly - that is it has to run in *"real time"*. There is usually a great deal of care and testing involved with programming for electronic products - while for some reason clients seem willing to accept bugs in windows software as somewhat normal, they are pretty unacceptable in, say, an engine control unit for a modern car.



The firmware may be developed by a separate team, and it is at the prototype stage that the two projects come together. (See: [More info - Firmware](#)) When the

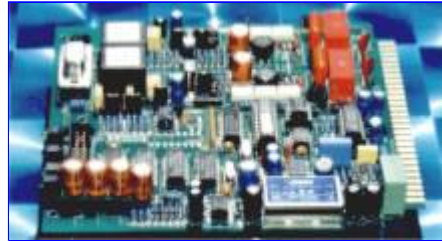
two components are integrated and prototype testing is complete, the finished project is handed over to the customer.

The trouble with doing something right the first time is that nobody appreciates how difficult it was. -- from "Fortunes" 😊

[Detail: Protos](#)[Detail: Firmware](#)[Example](#)[S/W](#)

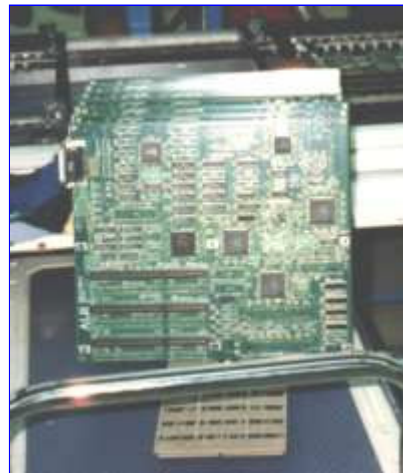
Pilot Run

To test the product further, a Pilot run normally follows the prototyping stage (See: [More info - PCB Assembly](#)). In the Pilot run, a small quantity of units (for instance 25) are field trialed in a beta test. The Pilot run is also an opportunity to assess the manufacturability of the design, and the useability of the documentation (See: [More info - PCB Documentation](#)).

[Detail: PCB Doc](#)[Detail: PCB Assem](#)[Photo](#)

Production

Following the pilot run there will likely be changes to the firmware, and possibly the circuit design, as the unit develops into a stable, final product. This process is controlled by [ECOs and version numbers](#). We handle the actual production of electronics for some customers, but others prefer to produce the circuitry in house, or use other assembly subcontractors. Our technical documentation is sufficiently detailed for our customers to have the electronics produced by a third party subcontractor. The **cost** of the final production, and to some degree the **style of design**, is heavily influenced by the **number of units** manufactured ([see Economies of scale](#)).

[Detail: PCB Assembly](#)

Innovation and Inspiration ... an opinion

For five seconds you're a **genius**, the rest of your life a bum.
Write it down. -- Lawrence Miller

Inspiration - Why have we left this until last? We all know it has to come first!
The **electronic design** process can be **methodical**, almost mechanical, but the

creativity at the start uses a very different set of human qualities. Throughout the high-tech business world the most amazing **success stories** are **of inspired people**, not large bureaucratic engineering departments with efficient Total Quality Management. (No offence intended - we work to standards ourselves)



Creative aptitude is really important in making new technology products **come to life**, and also in identifying the market and the need the product will fill. The ability to [bring that inspiration through](#) into a final product requires the level-headed decision making. Some electronic designs are not really new products as such, but developments of existing products. These designs, also, require **creative vision** to see which new features are really the most desirable, and how the design can be economically produced.

Bring in ideas and entertain them royally, for one of them may be king. --Mark Van Doren